

Assignment 1 – Cloud Deployment using Google Dataproc

This report documents the deployment of Assignment 1 on a cloud-based Hadoop environment using Google Cloud Dataproc. The objective was to demonstrate the execution of distributed data processing tasks using Apache Pig and Apache Hive on a managed Hadoop cluster.

1. Cloud Service Used

Google Cloud Platform (GCP) Dataproc was selected as the cloud service for this deployment. Dataproc is a managed service that provides Apache Hadoop, Pig, Hive, and YARN with minimal configuration overhead. As a new user, Google Cloud provides \$300 in free credits, which were sufficient for completing this assignment.

2. Environment Setup

A new GCP project was created and the required APIs were enabled which are highlighted on the lab doc webpage, including Compute Engine, Cloud Storage, Cloud Resource Manager, and Dataproc. IAM permissions were configured by assigning the Dataproc Worker role to the Compute Engine default service account to allow Dataproc agents to run correctly.

A Dataproc cluster was created in the europe-west1 region using the Google Cloud Console. The cluster consisted of one master node and two worker nodes running Hadoop, Pig, and Hive.

3. Data Storage and Preparation

The Spotify dataset (dataset.csv) and all Pig and Hive scripts were uploaded to a Google Cloud Storage (GCS) bucket. GCS was used instead of local storage or HDFS to simplify data access and avoid permission issues.

4. Pig Execution (Data Cleaning and Analysis)

The cleaning.pig script was submitted to the Dataproc cluster as a Pig job using the Dataproc Jobs interface. This script loaded the raw CSV dataset from GCS, removed headers, cast fields to appropriate data types, and stored the cleaned output back to GCS. As you can see in my screencast, this did not work for the first time around. I had to make changes which were specifically around hardcoded file names, I had to change my inputs and outputs to gs:// file location types. I realized that this would be applicable to all my code so therefore I had to change both my pig scripts and my hive script using this knowledge and re upload to my google cloud storage bucket.

After successful cleaning, the Analysis.pig script was executed to perform analytical queries such as filtering artists, identifying tracks with high danceability, and computing aggregate statistics. Both Pig jobs completed successfully, confirming correct distributed execution.

5. Hive Execution (Querying)

Hive queries were executed using a Hive job submission through the Dataproc console. An initial semantic error caused by using the LOCAL keyword with a GCS path was resolved by removing LOCAL and allowing Hive to read directly from Google Cloud Storage.

Once corrected, Hive tables were created successfully and queries were executed to analyse the cleaned dataset. Query results were verified using SELECT statements.

6. Monitoring and Verification

All Pig and Hive jobs were monitored using the Dataproc Jobs panel and the YARN ResourceManager web interface. These interfaces confirmed that the jobs were executed as distributed MapReduce applications on the cluster.

7. Screenshot and Cleanup

A 1–2 minute screencast was recorded showing the running Dataproc cluster, successful Pig and Hive jobs, YARN ResourceManager, and output data in Cloud Storage. The video was uploaded to Google Drive with public viewing access.

After completing the demonstration, the Dataproc cluster was deleted to avoid unnecessary billing.

8. Conclusion

This deployment demonstrated the successful execution of Apache Pig and Hive workflows on a managed cloud-based Hadoop platform. Google Dataproc provided an efficient and scalable environment for distributed data processing with minimal setup complexity.

Google drive link with screencast and screenshots of cluster work:

https://drive.google.com/drive/folders/1rFr2c1IKc0F4lx9vP24BuXPQnCOF_tau?usp=drive_link